

Sec 2.4 – Interpretations of the Derivative

(these notes are adapted from a variety of sources, including notes provided by Dr. Kristin Camenga)

Remember, the derivative of a function tells us:

- The slope of the tangent line to the function
- The instantaneous rate of change of the function

Both of these look at the change in the y value divided by the change in the x value. This means that *the units for the derivative are the units for y divided by the units for x* . We often write this in Leibniz notation as $\frac{dy}{dx}$

Examples:

- 1 For a time t in hours, $g(t)$ gives the temperature of a room in degrees Fahrenheit. What are the units of $g'(t)$?
- 2 For a pressure in atmospheres, $h(t)$ gives the volume of a gas in cubic meters. What are the units of $h'(t)$?

ANSWERS: 1) degrees Fahrenheit/hour 2) m^3/atm

Suppose $C(x)$ is the total cost for a company to produce x units of some commodity. Then C is called a cost function and the rate of change of C , $C'(x)$ is called the *marginal cost*.

Suppose that the cost, in dollars, for a company to produce x pairs of a new line of jeans is $C(x)$. Suppose $C'(100) = 11$.

What does this mean?

When producing the 100th pair of jeans, the production cost is increasing by \$11/pair.

How can you express the following statement as a derivative?

When producing the 500th pair of jeans, the production cost is increasing by \$6/pair.

$C'(500) = 6$

Recall that the inverse function f^{-1} switches inputs and outputs to the function.

That is, if $f(a) = b$, then $f^{-1}(b) = a$.

Suppose f is a function where the input is time in seconds and the output is meters. What are the meaning of the input and outputs to f^{-1} ?

Another way to think of the inverse is as a 'reverse lookup' function. For example, our function f answers the question *what is the position at a given time?*; f^{-1} reverses this question to answer *what is the time for a given position?*

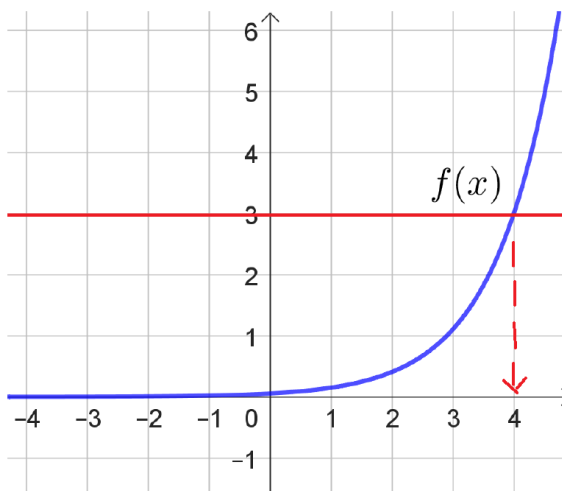
f returns distance, given an input of time

f^{-1} returns time, given an input of distance

Two different functions are given below. In each case, find $f^{-1}(3)$.

x	$f(x)$
0	7
1	5
2	3
4	1
5	-1

We read backward from the y -value of 3 to get the x -value of 2: $f^{-1}(3) = 2$.



We find where the graph intersects $y = 3$ and read off the x -value of 4: $f^{-1}(3) = 4$.

Partner up

The time for a chemical reaction, T (in minutes), is a function of the amount of catalyst present, a (in milliliters), so $T = f(a)$.

- If $f(5) = 18$, what are the units of 5? What are the units of 18?
What is the meaning of this statement?
 - If $f'(5) = -3$, what are the units of 5? What are the units of -3 ?
What is the meaning of this statement?
 - If $f^{-1}(8) = 10$, what are the units of 8? What are the units of 10?
What is the meaning of this statement?
 - If $(f^{-1})'(8) = 4$, what are the units of 8? What are the units of 4?
What is the meaning of this statement?
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- The units of 5 are milliliters (of catalyst) and for 18 are minutes (of time for the reaction). So if we have 5 mL of catalyst, the reaction takes 18 minutes.
 - The units of 5 are milliliters (of catalyst) and for -3 are minutes/milliliter - the time is decreasing by 3 min/mL as we add more catalyst to the 5mL we have. (This says that the more catalyst, the less time for the reaction.)
 - If $f^{-1}(8) = 10$, the units of 8 are minutes and the units of 10 are mL. This statement says that if we want the reaction to take 8 minutes, we need 10mL of catalyst.
 - If $(f^{-1})'(8) = 4$, the units of 8 are minutes and the units of 4 are mL/min. The statement says that when the reaction is taking 8 minutes, we need 4 mL of catalyst per minute longer of reaction.